

LEVEL-I (MAIN)

EXERCISE-I

Straight Objective Type Questions

Current, drift velocity, Ohm's Law

1. An electron of mass m , moves around the nucleus in a circular orbit of radius ' r ' under the action of centripetal force ' F '. The equivalent electric current is

1) $\frac{e}{2\pi} \sqrt{\frac{F}{mr}}$ 2) $2\pi e \sqrt{\frac{F}{mr}}$ 3) $\frac{e}{\pi} \sqrt{\frac{F}{mr}}$ 4) $\frac{e}{2\pi} \sqrt{\frac{mr}{F}}$

2. A copper conductor of area of cross-section 40 mm^2 on a side carries a constant current of $32 \times 10^{-6} \text{ A}$. Then the current density is (in amp/m^2)

1) 1.6 2) 0.8 3) 0.4 4) 3.2

3. Positive and negative ions are produced in the atmosphere due to cosmic rays from space and also due to radioactive elements in the soil. In some region in the atmosphere, the electric field strength is 100 V/m in the vertically downward direction. This field exerts force on the positive and negative ions in the given region in atmosphere. As a result, positive ions, having a density $500/\text{cm}^3$ drift downward while negative ions, having a density $300/\text{cm}^3$ drift upward. All these ions are singly charged. It is observed that the conductivity in the given region is $4 \times 10^{-13} (\Omega\text{-m})^{-1}$. Find the average speed of ions, assuming it to be the same for positive and negative ions.

1) 1.3 m/s 2) 0.31 m/s 3) 0.93 m/s 4) 1.6 m/s

4. Electric current through a conductor varies with time as $I(t) = 50 \sin(100\pi t)$. Here I is in amperes and t in seconds. Total charge that passes any point from $t = 0$ to $t = \frac{1}{200} \text{ s}$ is

1) 1.2 C 2) 0.36 C 3) 0.16 C 4) 0.02 C

5. Drift speed of electrons, when 1.5 A of current flows in a copper wire of cross section 5 mm^2 , is v . If the electron density in copper is $9 \times 10^{28}/\text{m}^3$ the value of v in mm/s is close to (Take charge of electron to be $= 1.6 \times 10^{-19}$)

1) 0.2 2) 3 3) 2 4) 0.02

6. A current of 5A passes through a copper conductor (resistivity $= 1.7 \times 10^{-8} \Omega\text{m}$) radius of cross section 5mm. Find the mobility of the charges if their drift velocity is $1.1 \times 10^{-3} \text{ m/s}$.

1) $1.8 \times \text{m}^2/\text{Vs}$ 2) $1.5 \times \text{m}^2/\text{Vs}$ 3) $1.0 \times \text{m}^2/\text{Vs}$ 4) $1.3 \times \text{m}^2/\text{Vs}$

Resistance, Effective Resistance, Variation of Resistance with temperature

7. The resistance of the wire is 121 ohm. It is divided into ' n ' equal parts and they are connected in parallel, then effective resistance is 1 ohm. The value of ' n ' is

1) 12 2) 13 3) 11 4) 3

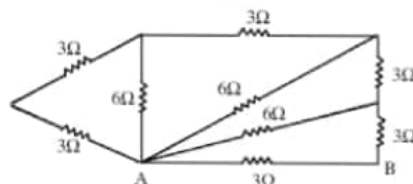
8. The resultant resistance of two resistance in series is 50Ω and it is 12Ω , when they are in parallel. The individual resistances are

1) 20Ω and 15Ω 2) 15Ω and 30Ω
3) 20Ω and 30Ω 4) 10Ω and 15Ω

9. A wire of resistance 3Ω is stretched uniformly then the length is doubled. The wire is now bent in the form of an equilateral triangle then the effective resistance between the ends of any side of the triangle

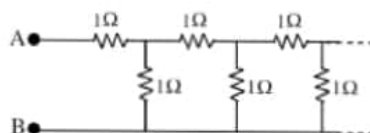
- 1) $\frac{9}{2}\Omega$ 2) $\frac{8}{3}\Omega$ 3) 2Ω 4) 1Ω

10. The effective resistance between A and B is the given circuit is



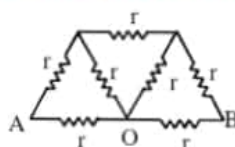
- 1) 2Ω 2) 3Ω 3) 9Ω 4) 6Ω

11. The equivalent resistance between points A and B of an infinite network of resistance each of 1Ω connected as shown is



- 1) infinite 2) 2Ω 3) zero 4) $\left(\frac{1+\sqrt{5}}{2}\right)$

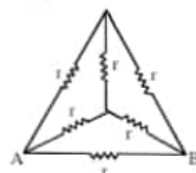
12. Equivalent resistance across A and B in the given circuit is



- 1) $7r$ 2) $8r/7$ 3) $4r/7$ 4) $7r/8$

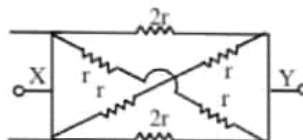
13. Effective resistance between A and B in the given circuit is

- 1) $6r$
2) r
3) $2r$
4) $r/2$



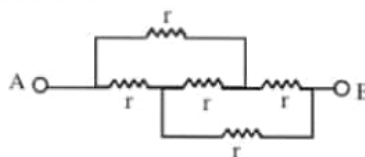
14. The equivalent resistance across XY is

- 1) r
2) $2r$
3) $r/2$
4) $r/4$



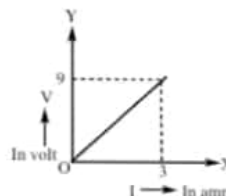
15. Effective resistance between A and B in the given circuit is

- 1) r
- 2) $3r$
- 3) $5r$
- 4) $4r$



16. The variation of current and voltage in a conductor has been shown in figure. The resistance of the conductor is

- 1) 4Ω
- 2) 2Ω
- 3) 3Ω
- 4) 1Ω



17. An aluminium ($\alpha_{Al} = 4 \times 10^{-3} / ^\circ\text{C}$) wire resistance ' R_1 ' and carbon wire ($\alpha_c = 0.5 \times 10^{-3} / ^\circ\text{C}$) resistance ' R_2 ' are connected in series to have a resultant resistance of 18 ohm at all temperatures. The values of R_1 and R_2 in ohms

- 1) 2, 16
- 2) 12, 6
- 3) 13, 5
- 4) 14, 4

18. Two resistors with temperature coefficients of resistance α_1 and α_2 have resistances R_1 and R_2 at 0°C . Find the temperature coefficient of the compound resistor consisting of the two resistors connected in parallel

- 1) $\frac{R_1\alpha_1 + R_2\alpha_2}{R_1 + R_2}$
- 2) $\frac{\alpha_1 + \alpha_2}{2}$
- 3) $\frac{2\alpha_1\alpha_2}{\alpha_1 + \alpha_2}$
- 4) $\frac{R_1\alpha_2 + R_2\alpha_1}{R_1 + R_2}$

19. A copper wire is stretched to make it 0.5% longer. The percentage change in its electrical resistance if its volume remains unchanged is

- 1) 2.5%
- 2) 0.5%
- 3) 1.0%
- 4) 2.0%

20. A carbon resistance has a following colour code. What is the value of the resistance ?



- 1) $1.64 M\Omega \pm 5\%$
- 2) $530 k\Omega \pm 5\%$
- 3) $64 k\Omega \pm 10\%$
- 4) $5.3 M\Omega \pm 5\%$

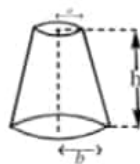
21. A uniform metallic wire has a resistance of 18Ω and is bent into an equilateral triangle. Then, the resistance between any two vertices of the triangle is

- 1) 4Ω
- 2) 8Ω
- 3) 12Ω
- 4) 2Ω

22. A thick spherical shell has inner radius a and outer b . The material has resistivity ρ . When a potential difference is applied between the inner and outer surfaces, its resistance is

- 1) $\frac{\rho}{4\pi} \frac{ab}{(b-a)}$
- 2) $\frac{\rho(b-a)}{4\pi ab}$
- 3) $\frac{4\pi ab}{\rho(b-a)}$
- 4) $\frac{\rho(b-a)}{2\pi ab}$

23. A material of resistivity ρ is formed in the shape of a truncated cone of altitude h . The top end has a radius a while bottom b . Assuming a uniform current density through any circular cross-section of the cone. Find the resistance between the two ends.



- a) $\frac{\rho}{2\pi} \frac{h}{ab}$ b) $\frac{2\rho}{\pi} \frac{h}{ab}$ c) $\frac{\rho}{\pi} \frac{h}{ab}$ d) $\frac{\rho}{\pi} \frac{h}{(a+b)}$
24. The cross-section area and length of cylindrical conductor are A and l respectively. The specific conductivity varies as $\sigma(x) = \sigma_0 \frac{\ell}{\sqrt{x}}$, where x is the distance along the axis of the cylinder from one of its ends. Compute the resistance of the system along the cylindrical axis.
- a) $\frac{3\sqrt{\ell}}{2A\sigma_0}$ b) $\frac{2\ell}{3A\sigma_0}$ c) $\frac{2\sqrt{\ell}}{3A\sigma_0}$ d) $\frac{\sqrt{\ell}}{A\sigma_0}$

Cells, Grouping of cells

25. The emf of a storage battery is 90 V before charging and 100 V after charging. When charging began the current was 10 A. What is the current at the end of charging if the internal resistance of the storage battery during the whole process of charging may be taken as constant and equal to 2 ohms ?
1) 5 A 2) 10 A 3) 15 A 4) 20 A
26. How will you connect (series and parallel) 24 cells each of internal resistance 1 ohm so as to get maximum power output across a load of 10Ω ? [a=number of rows that are connected in parallel b=number of cells in series in each row]
1) a = 6, b = 4 2) a = 8, b = 3 3) a = 2, b = 12 4) a = 1, b = 24
27. Fig. shows a part of circuit. When a current $I = 1.5\text{A}$ passes through it in the direction shown, power delivered to it (section XY) is 75W. Potential difference between X and Y and also the emf E of the cell are respectively (assume that the cell has a negligible internal resistance)



- 1) 50 V, 44 V 2) 50 V, 48 V 3) 36 V, 50 V 4) 48 V, 50 V
28. A battery consists of a variable number n of identical cells having internal resistance connected in series. The terminals of the battery are short circuited and the current I measured. Which one of the graph below shows the relationship between I and n ?



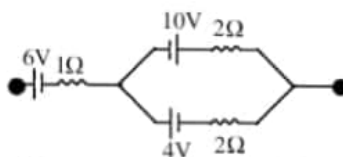
29. In previous problem, if the cell had been connected in parallel (instead of in series) which of the above graphs would have shown the relationship between total current I and n ?



30. A cell of emf E has an internal resistance r & is connected to rheostat. When resistance R of rheostat is changed correct graph of potential difference across it is



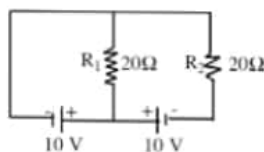
31. Find the emf and internal resistance of a single battery which is equivalent to a combination of three batteries as shown in figure.



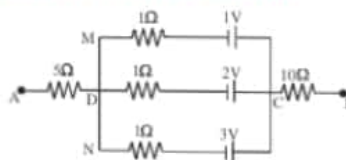
- 1) 3V, 5Ω 2) 3V, 2Ω 3) 7V, 2Ω 4) 20V, 5Ω

32. In the given circuit the cells have zero internal resistance. The currents (in amperes) passing through resistance R_1 and R_2 respectively, are

- 1) 1, 2
2) 2, 2
3) 0.5, 0
4) 0, 1

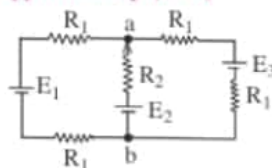


33. In the circuit, the potential difference between A and B is



- 1) 6 V 2) 1 V 3) 3 V 4) 2 V

34. For the circuit shown, with $R_1=1.0\Omega$, $R_2=2.0\Omega$, $E_1=2\text{ V}$ and $E_2=E_3=4\text{ V}$, the potential difference between the points 'a' and 'b' is approximately (in V)



- 1) 3.3 2) 2.7 3) 3.7 4) 2.3

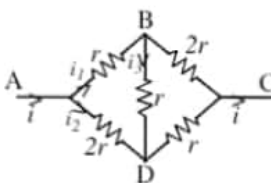
35. A cell of internal resistance r drives current through an external resistance R . The power delivered by the cell to the external resistance will be maximum when

1) $R = 1000 r$ 2) $R = r$ 3) $R = 2r$ 4) $R = 0.001 r$

Kirchhoff's laws and applications

36. The value of current i_j in the given circuit is

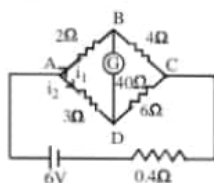
1) $\frac{i}{5}$
2) $\frac{2i}{5}$
3) $\frac{3i}{5}$ 4) $\frac{4i}{5}$



37. A balanced wheatstone bridge is shown the values of currents i_1 and i_2 are

(ATB)

1) 0.9A, 0.6A
2) 0.3A, 0.2A
3) 0.5A, 0.7A
4) 0.7A, 0.6A



38. A metallic conductor at 10°C connected in the left gap of Meter Bridge gives balancing length 40cm. When the conductor is at 60°C , the balancing point shifts by ___ cm, (temperature coefficient of resistance of the material of the wire is $(1/220)^\circ\text{C}^{-1}$)

1) 4.8 2) 8.7 3) 15 4) 7

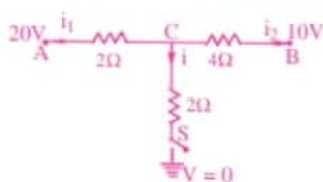
39. Two unknown resistances X and Y are connected to left and right gaps of a meter bridge and the balancing point is obtained at 80cm from left. When a 10Ω resistance is connected parallel to X , the balancing point is 50 cm from left. The values of X and Y respectively are

1) $40\Omega, 9\Omega$ 2) $30\Omega, 7.5\Omega$ 3) $20\Omega, 6\Omega$ 4) $10\Omega, 3\Omega$

40. In a meter bridge, the balance length from left end (standard resistance of 1Ω is in the right gap) is found to be 20cm. The length of Resistance in left gap is $\frac{1}{2}\text{m}$ and radius is 2mm. Its specific resistance is

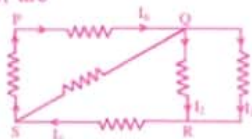
1) $\pi \times 10^{-6} \text{ ohm-m}$ 2) $2\pi \times 10^{-6} \text{ ohm-m}$ 3) $\frac{\pi}{2} \times 10^{-6} \text{ ohm-m}$ 4) $3\pi \times 10^{-6} \text{ ohm-m}$

41. When the switch S , in the circuit shown, is closed, then the value of current i will be:



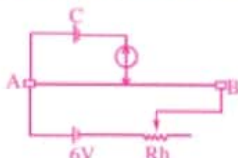
1) 3A 2) 5A 3) 4A 4) 2A

42. In the given circuit diagram, the currents, $I_1 = -0.3$ A, $I_4 = 0.8$ A and $I_5 = 0.4$ A, are flowing as shown. The currents I_2 , I_3 and I_6 , respectively, are



- 1) 1.1 A, -0.4 A, 0.4 A 2) 1.1 A, 0.4 A, 0.4 A 3) 0.4 A, 1.1 A, 0.4 A 4) -0.4 A, 0.4 A, 1.1 A
43. The resistance of the meter bridge AB is given figure is 4Ω . With a cell of emf $\mathcal{E}_1 = 0.5$ V and rheostat resistance $R_h = 2\Omega$ the null point is obtained at some point J. When the cell is replaced by another one of emf $\mathcal{E} = \mathcal{E}_2$ the same null point J is found for $R_h = 6\Omega$. The emf $\mathcal{E}_2$ is

- 1) 0.6 V
2) 0.5 V
3) 0.3 V
4) 0.4 V



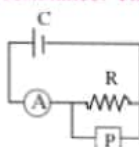
Potentiometer, Ammeter and voltmeter

44. The scale of a galvanometer is divided into 150 equal divisions. The galvanometer has the current sensitivity of 10 divisions per mA and the voltage sensitivity of 2 divisions per mV. How the galvanometer be designed to read i) 6 A, per division and ii) 1 V, per division ?

- 1) $40\mu\Omega$ is connected in parallel, 1000Ω is connected in series
2) $20\mu\Omega$ is connected in parallel, 9995Ω is connected in series
3) $8.3\mu\Omega$ is connected in parallel, 9995Ω is connected in series
4) $8.3 \times 10^{-5}\Omega$ is connected in parallel, 9995Ω is connected in series

45. An ammeter A of finite resistance, and a resistor R are joined in series to an ideal cell C. A potentiometer P is joined in parallel to R. The ammeter reading is I_0 and the potentiometer reading is V_0 . P is now replaced by a voltmeter of finite resistance. The ammeter reading now is I and the voltmeter reading is V.

- 1) $I > I_0$, $V < V_0$
2) $I > I_0$, $V = V_0$
3) $I = I_0$, $V < V_0$ 4) $I < I_0$, $V = V_0$



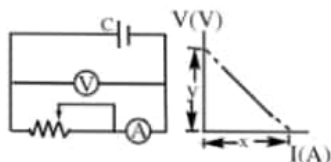
46. An ammeter is connected to measure the current intensity in a circuit with a resistance R. What relative error will be made if connection of the ammeter does not change the current intensity in the circuit? The voltage across the ends of the circuit is kept constant.

- 1) $\frac{R}{R_0}$ 2) $\frac{R_0}{R}$ 3) $\frac{1}{1 + \frac{R}{R_0}}$ 4) $\frac{1}{1 + \frac{R_0}{R}}$

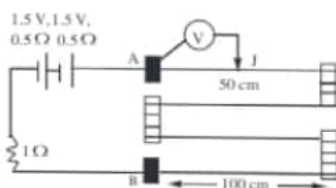
47. The voltage across a resistance R is determined using a voltmeter connected to its ends. What relative error will be made if the readings of the voltmeter are taken as the voltage applied before it was switched on ? The current intensity in the circuit is constant.

- 1) $\frac{R}{R_0}$ 2) $\frac{R_0}{R}$ 3) $\frac{1}{1 + \frac{R}{R_0}}$ 4) $\frac{1}{1 + \frac{R_0}{R}}$

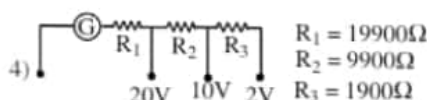
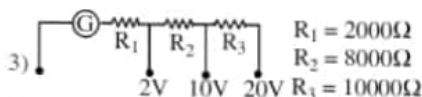
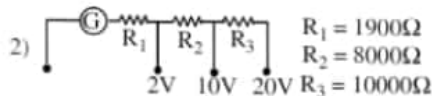
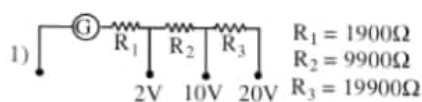
48. The diagram besides shows a circuit used in an experiment to determine the emf and internal resistance of the cell C. A graph was plotted of the potential difference V between the terminals of the cell against the current I , which was varied by adjusting the rheostat. The graph is shown on the right; x and y are the intercepts of the graph with the axes as shown. What is the internal resistance of the cell?



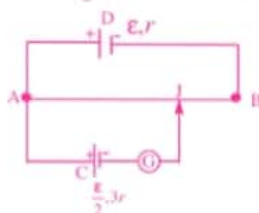
- 1) x 2) y 3) x/y 4) y/x
49. A moving coil galvanometer has a coil with 175 turns and area 1cm^2 . It uses a torsion band of torsion constant 10^{-6}N-m/rad . The coil is placed in a magnetic field B parallel to its plane. The coil deflects by 1° for a current of 1mA . The value of B (in Tesla) is approximately:
- 1) 10^{-2} 2) 10^{-1} 3) 10^{-3} 4) 10^{-4}
50. The resistance of a galvanometer is 50 ohm and the maximum current which can be passed through it is 0.002 A . What resistance must be connected to it in order to convert it into an ammeter of range $0 - 0.5\text{A}$?
- 1) 0.5 ohm 2) 0.02 ohm 3) 0.2 ohm 4) 0.002 ohm
51. In the circuit shown, a four-wire potentiometer is made of a 400 cm long wire, which extends between A and B. The resistance per unit length of the potentiometer wire is $r = 0.01\text{ }\Omega/\text{cm}$. If an ideal voltmeter is connected as shown with jockey J at 50 cm from end A, the expected reading of the voltmeter will be



- 1) 0.50 V 2) 0.75 V 3) 0.20 V 4) 0.25 V
52. A galvanometer of resistance $100\text{ }\Omega$ has 50 divisions on its scale and has sensitivity of $20\text{ }\mu\text{A/division}$. It is to be converted to a voltmeter with three ranges, of $0-2\text{V}$, $0-10\text{ V}$ and $0-20\text{V}$. The appropriate circuit to do so is

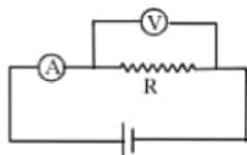


53. A potentiometer wire AB having length L and resistance $12r$ is joined to a cell D of emf ε and internal resistance r . A cell C having emf $\varepsilon/2$ and internal resistance $3r$ is connected. The length AJ at which the galvanometer as shown in figure shows no deflection is:



- 1) $\frac{11}{12}L$ 2) $\frac{11}{24}L$ 3) $\frac{13}{24}L$ 4) $\frac{5}{12}L$

54. The actual value of resistance R , shown in the figure is 30Ω . This is measured in an experiment as shown using the standard formula $R = \frac{V}{I}$, where V and I are the readings of the voltmeter and ammeter, respectively. If the measured value of R is 5% less, then the internal resistance of the voltmeter is

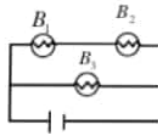


- 1) 600Ω 2) 570Ω 3) 35Ω 4) 350Ω

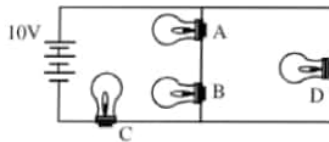
Electric power, Joule's law

55. A 100 watt immersion heater is placed in a pot containing 1 litre of water at 20°C . How long will it take to heat water to boiling temperature if 20% of the available energy is lost to surroundings?
- 1) 60 minutes 2) 70 minutes 3) 44 minutes 4) 32 minutes
56. The same mass of copper is drawn into two wires 1 mm thick and 2 mm thick. Two wires are then connected in series and current is passed. The heat produced in the wires is the ratio
- 1) 16 : 1 2) 1 : 1 3) 4 : 1 4) 9 : 1
57. A long resistance wire is divided into $2n$ parts. Then n parts are connected in series and other ' n ' parts in parallel separately. Both combinations are connected to identical supplies. Then the ratio of heat produced in series to parallel combinations will be
- 1) 1 : 1 2) 1 : n^2 3) 1 : n^4 4) n^2 : 1
58. A resistance coil of 60 ohm is immersed in 42000 gm of water. A current of 7A is passed through it. Calculate the rise in temperature per minute. The sp. heat of water = $4200 \frac{\text{J}}{\text{kg} \cdot ^\circ\text{K}}$
- 1) 5°C 2) 3°C 3) 2°C 4) 1°C
59. The maximum voltage that can be applied to $5\text{K}\Omega$ and 8W resistor without exceeding its heat dissipating capacity is
- 1) 100 V 2) 400 V 3) 160 V 4) 200 V

60. A 100 W bulb B_1 and two 60 W bulbs B_2 and B_3 are connected to a 250 V source as shown. Now W_1 , W_2 and W_3 are the output powers of the bulbs B_1 , B_2 and B_3 respectively, then :



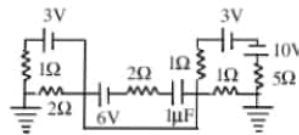
- 1) $W_1 > W_2 = W_3$ 2) $W_1 > W_2 > W_3$ 3) $W_1 < W_2 = W_3$ 4) $W_1 < W_2 < W_3$
61. In the circuit diagram, all the bulbs are identical. Which bulb will be the brightest?



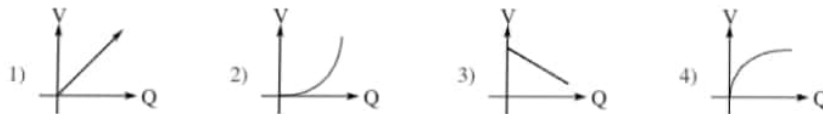
- 1) A 2) B 3) C 4) D

R-C circuits

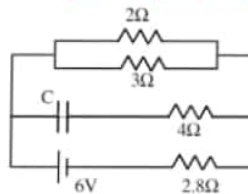
62. For the circuit shown in the figure, determine the charge of capacitor in steady state.



- 1) $4\mu\text{C}$ 2) $6\mu\text{C}$ 3) $1\mu\text{C}$ 4) Zero
63. A capacitor is charged steadily from a DC source. Correct variation of potential difference across the plates of capacitor with charge on the plates of capacitor is



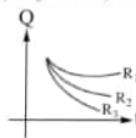
64. In the steady state, the charge on the capacitor capacity $0.2\mu\text{F}$ is



- 1) zero 2) $0.36\mu\text{C}$ 3) $0.2\mu\text{C}$ 4) $1.2\mu\text{C}$

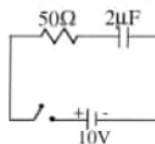
65. Three identical capacitors are charged by connecting them in parallel across a battery of V volt. They are then allowed to discharge via resistor R_1 , R_2 and R_3 . Here, Q charge versus t time graphs are shown. Then,

- 1) R_1 is smallest 2) R_2 is smallest
3) R_3 is smallest 4) $R_3 > R_2 < R_1$

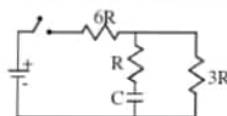


66. For the circuit shown in figure below, at $t = 0$, switch is closed, the initial current through resistor and final charge on capacitor are

- 1) $5A$, $5\mu C$ 2) $5A$, $20\mu C$
3) $0.2A$, $5\mu C$
4) $0.2A$, $20\mu C$

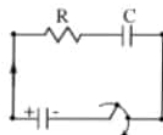


67. For the given circuit shown in figure below, time constant is

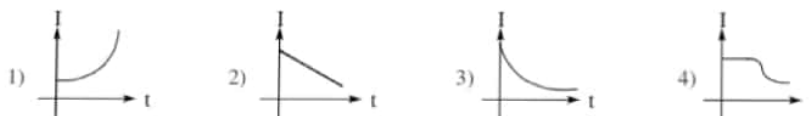


- 1) $\tau = RC$ 2) $\tau = 2.1RC$ 3) $\tau = 27/4RC$ 4) $\tau = 3RC$

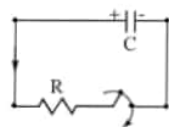
68. During charging of capacitor in the circuit shown,



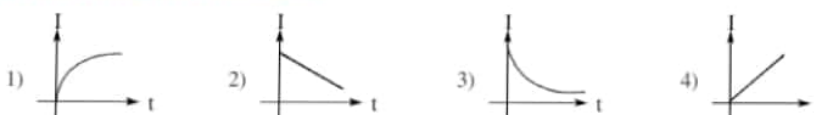
Circuit current versus time graph is



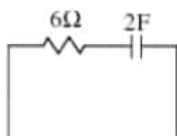
69. During discharging of a capacitor via a resistor



Circuit current varies with time as



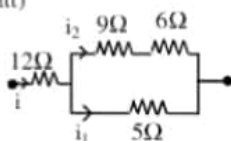
70. In a capacitor shown in the circuit is charged to 5V and left in the circuit in 12s, the charge on the capacitor will become



- 1) $\frac{10}{e}C$ 2) $\frac{e}{10}C$ 3) $\frac{10}{e^2}C$ 4) $\frac{e^2}{10}C$ ($e = 2.718$)

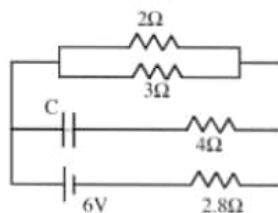
Numerical Value type Questions

71. Electric current through a conductor varies with time as $I(t) = 50 \sin(100\pi t)$. Here I is in amperes and t in seconds. Total charge that passes any point from $t = 0$ to $t = \frac{1}{200}$ s is (in coulombs)
72. The current in a conductor varies with time ' t ' as $I = 3t + 4t^2$. Where I in amp and t in sec. The electric charge flows through the section of the conductor between $t = 1$ s and $t = 3$ s (in coulombs)
73. A carbon filament has resistance of 120Ω at 0°C what must be the resistance of a copper filament connected in series with carbon so that combination has same resistance at all temperatures (in Ω) ($\alpha_{\text{carbon}} = 5 \times 10^{-4}/^\circ\text{C}$, $\alpha_{\text{copper}} = 4 \times 10^{-3}/^\circ\text{C}$)
74. The resistance of an Carbon wire is 10Ω . A current of 30 mA is flowing in it at 20°C . If p.d. across its ends constant, then its temperature is increased to 120°C then the current flowing in the wire will be in mA ($\alpha_c = 5 \times 10^{-3}/^\circ\text{C}$)
75. A cell whose e.m.f. is 2V and internal resistance is 0.1Ω is connected with a resistance of 3.9Ω the voltage across the cell terminal will be ---- Volt.
76. A metallic conductor at 10°C connected in the left gap of Meter Bridge gives balancing length 40cm. When the conductor is at 60°C , the balancing point shifts by ___ cm, (temperature coefficient of resistance of the material of the wire is $(1/220)/^\circ\text{C}$)
77. A moving coil galvanometer of resistance 20Ω gives a full scale deflection when a current of 1 mA is passed through it. It is to be converted into an ammeter reading 20 A on full scale. But the shunt of 0.005Ω only is available. What resistance should be connected in series with the galvanometer coil ? (in Ω)
78. In the following circuit, 5Ω resistor develops 45J/s due to current flowing through it. The power developed across 12Ω resistor is (in watt)



79. Two wires A and B of same material and mass have their lengths in the ratio 1 : 2. On connecting them separately to the same source, the rate of heat dissipation in B is found to be 5W. The rate of heat dissipation in A is

80. In the steady state, the charge on the capacitor capacity $0.2\mu\text{F}$ is (in μC)



EXERCISE-II

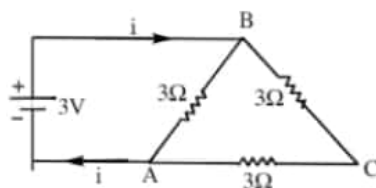
Current Drift velocity, Ohm's law

1. A current of 4.8 A is flowing in a conductor the number of electrons passing per sec through the conductor will be

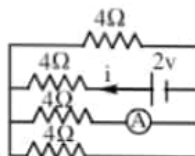
- 1) 3×10^{19} 2) 7.68×10^{20} 3) 76.8×10^{19} 4) 3×10^{20}

2. The current 'i' in the circuit as shown in fig. is

- 1) 2A
2) $\frac{1}{3}\text{A}$
3) 1A
4) 1.5 A



3. Current coming from the battery and ammeter reading are

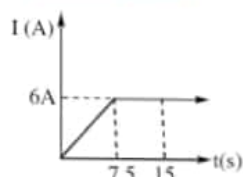


- 1) $\frac{3}{8}\text{A}, \frac{1}{8}\text{A}$ 2) $\frac{1}{8}\text{A}, \frac{1}{8}\text{A}$ 3) $2\text{A}, \frac{2}{3}\text{A}$ 4) $2\text{A}, \frac{1}{8}\text{A}$

4. A uniform copper wire carries a current i amperes and has p carriers per metre³. The length of the wire is l metre and its cross-section area is s metre². If the charge on a carrier is q coulombs, the drift velocity in ms^{-1} is given by

- 1) i/lsq 2) i/psq 3) psq/i 4) $i/pslq$

5. The graph of current versus time in a wire is given by

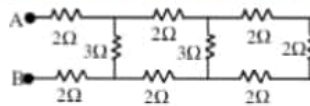


If charge that flows through a wire in 7.5 s is q_1 and that flows through the wire in 15s is q_2 , then ratio $q_1 : q_2$ is

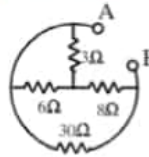
- 1) 2 : 1 2) 1 : 2 3) 4 : 1 4) 1 : 4

Resistance, effective Resistance, variation of resistance with temperature

6. A copper wire of cross-sectional area 3.4 mm^2 and length of the wire 400 m , specific resistivity of copper is $1.7 \times 10^{-8} \Omega\text{-m}$. Then the resistance of the wire is
 1) 20Ω 2) 200Ω 3) 2Ω 4) $2 \text{ k}\Omega$
7. Four wires made of same material have different lengths and radii, the wire having more resistance in the following case is
 1) $\ell = 100 \text{ cm}, r = 1 \text{ mm}$ 2) $\ell = 50 \text{ cm}, r = 2 \text{ mm}$ 3) $\ell = 100 \text{ cm}, r = \frac{1}{2} \text{ mm}$ 4) $\ell = 50 \text{ cm}, r = \frac{1}{2} \text{ mm}$
8. The effective resistance in series combination of two equal resistance is 's'. When they are joined in parallel the total resistance is p. If $s = np$ then the minimum possible value of 'n' is
 1) 4 2) 1 3) 2 4) 3
9. Three resistances $1 \Omega, 2 \Omega$ and 3Ω are connected in parallel. The ratio of currents passing through them when a p.d. is applied across the ends is
 1) $2 : 3 : 6$ 2) $1 : 2 : 3$ 3) $1 : 1 : 1$ 4) $6 : 3 : 2$
10. You are given several identical resistors each of value 5Ω and each capable of carrying a maximum current of 2 A . It is required to make a suitable combination of these resistances to produce a resistance of 2.5Ω which can carry current of 4 A . The minimum number of resistances required for this job is
 1) 2 2) 4 3) 6 4) 8
11. The effective resistance between A and B in the given circuit is



- 1) 20Ω 2) 7Ω 3) 3Ω 4) 6Ω
12. The effective resistance between A and B in the given circuit is
 1) 1.5Ω
 2) 7.5Ω
 3) 4.7Ω
 4) 12Ω



13. The effective resistance between A and B in the given circuit is



- 1) 18Ω 2) 6Ω 3) 2Ω 4) 3Ω
14. A hollow cylinder ($\rho = 2.2 \times 10^{-8} \Omega\text{-m}$) of length 3 m has inner and outer diameter as 2 mm and 4 mm respectively. The resistance of the cylinder is
 1) $0.35 \times 10^{-3} \Omega$ 2) $3 \times 10^{-3} \Omega$ 3) $7 \times 10^{-3} \Omega$ 4) $3.1 \times 10^{-3} \Omega$
15. A metal block has length 50 cm , breadth 30 cm and thickness 20 cm . When current passes through it parallel to its length and parallel to its breadth, the ratio of maximum to minimum resistance of that block is
 1) $4 : 25$ 2) $25 : 4$ 3) $25 : 9$ 4) $9 : 25$

16. Two wires of equal diameters of resistivities ρ_1 and ρ_2 and length X_1 and X_2 respectively are joined in series. The equivalent resistivity of the combination is

1) $\frac{\rho_1 X_1 + \rho_2 X_2}{X_1 + X_2}$ 2) $\frac{\rho_1 X_2 + \rho_2 X_1}{X_1 - X_2}$ 3) $\frac{\rho_1 X_2 + \rho_2 X_1}{X_1 + X_2}$ 4) $\frac{\rho_1 X_1 + \rho_2 X_2}{X_1 - X_2}$

17. The resistance of a bulb filament is 100Ω at a temperature of 100°C . If its temperature coefficient of resistance be $0.005 \text{ per } ^\circ\text{C}$, its resistance will become 200Ω at a temperature of

1) 300°C 2) 400°C 3) 500°C 4) 200°C

18. **Statement-1 :-** The temperature dependence of resistance is usually given as $R = R_0(1 + \alpha\Delta t)$. The resistance of a wire changes from 100Ω to 150Ω when its temperature is increased from 27°C to 227°C . This implies that $\alpha = 2.5 \times 10^{-3} / ^\circ\text{C}$

Statement-2 :- $R = R_0(1 + \alpha\Delta t)$ is valid only when the change in the temperature ΔT is small and $\Delta R = (R - R_0) \ll R_0$.

- 1) Statement-1 is true, statement-2 is true, Statement-2 is the correct explanation of statement-1.
 2) Statement-1 is true, statement-2 is true, Statement-2 is not the correct explanation of statement-1
 3) Statement-1 is false, Statement-2 is true
 4) Statement-1 is true, Statement-2 is false

Cells, Grouping of cells

19. A 50 V battery is connected across a 10 ohm resistor. The current is 4.5 A is flowing through the resistor then the internal resistance of the battery is

1) zero 2) 0.5Ω 3) 1.1Ω 4) 5Ω

20. A cell whose e.m.f. is 2 V and internal resistance is 0.1Ω is connected with a resistance of 3.9Ω the voltage across the cell terminal will be

1) 0.5 V 2) 1.9 V 3) 1.95 V 4) 72 V

21. The emf of a Daniel cell is 1.08 V . When the terminals of the cells are connected to a resistance of 3Ω , the potential difference across the terminals is found to be 0.6 V . Then the internal resistance of the cell is

1) 1.8Ω 2) 2.4Ω 3) 3.24Ω 4) 0.2Ω

22. A student is asked to connected four cells of e.m.f. of 1 V and internal resistance 0.5 ohm in series with an external resistance of 1 ohm . But one cell is wrongly connected by him with its terminal reversed, the current in the circuit is

1) $\frac{1}{3} \text{ A}$ 2) $\frac{2}{3} \text{ A}$ 3) $\frac{3}{4} \text{ A}$ 4) $\frac{4}{3} \text{ A}$

23. When two identical cells are connected either in series or in parallel across a 4 ohm resistor, they send the same current through it. The internal resistance of the cell in ohm is

1) 1.2 2) 2 3) 4 4) 4.8

24. Two cells with same e.m.f. 'E' and different internal resistances r_1 and r_2 are connected in series to an external resistance 'R'. The value of R so that the p.d. across the first cell be zero is

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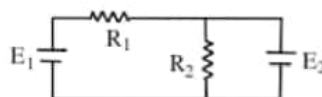
1) $\sqrt{r_1 r_2}$ 2) $r_1 + r_2$ 3) $r_1 - r_2$ 4) $\frac{r_1 + r_2}{2}$

25. The current in a circuit containing a battery connected to 2Ω resistance is 0.9A . When a resistance of 7Ω is connected to the same battery, the current observed in the circuit is 0.3A . Then the internal resistance of the battery.

1) 0.1Ω 2) 0.5Ω 3) 1Ω 4) zero

26. Two cells of emf E_1 and E_2 are connected to two resistors R_1 and R_2 as shown. If E_2 is short circuited then currents through R_1 and R_2 are

1) $\frac{E_1}{R_2}, 0$ 2) $0, \frac{E_1}{R_2}$
3) $\frac{E_1}{R_1}, 0$ 4) $0, \frac{E_1}{R_1}$

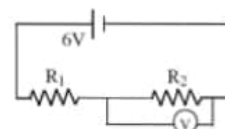


27. When two cells of different emf's are connected in series to an external resistance the current is 5A . When the poles of one cell are interchanged, the current is 3A . The ratio of emf's of two cell is

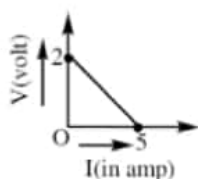
1) $2 : 1$ 2) $1 : 3$ 3) $5 : 1$ 4) $4 : 1$

28. In the circuit $R_1 = 400\text{ ohm}$ and $R_2 = 200\text{ ohm}$. If the resistance of the voltmeter $R = 10,000\text{ ohm}$, voltmeter reading is

1) 2V
2) greater
3) greater than 3V
4) less than 2V



29. For a cell, the graph between the p.d.(V) across the terminals of the cell and the current I drawn from the cell is shown in the fig. the emf and the internal resistance of the cell is E and r respectively.



1) $E = 2\text{ V}, r = 0.5\Omega$ 2) $E = 2\text{V}, r = 0.4\Omega$ 3) $E > 2\text{ V}, r = 0.5\Omega$ 4) $E > 2\text{ V}, r = 0.4\Omega$

30. How many cells each marked $(6\text{V} - 12\text{A})$ should be connected in mixed grouping so that it may be marked $(24\text{V} - 24\text{ A})$

1) 4 2) 8 3) 12 4) 6

31. The minimum number of cells in mixed grouping required to produce a maximum current of 1.5 A through an external resistance of 30Ω , given the emf of each cell is 1.5 V and internal resistance is 1Ω is

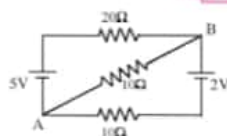
1) 30 2) 120 3) 40 4) 60

32. Three equal resistance each of 3Ω are in series and connected to cell of internal resistance one ohm. If three resistances are parallel and connected to the same cell. Then the ratio of the respective currents through the electric circuits in the two cases is

1) $1/8$ 2) $1/7$ 3) $1/5$ 4) $1/3$

33. The P.d between the terminals A & B is ____

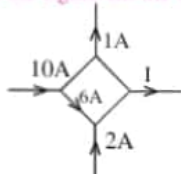
- 1) 2V
- 2) 3V
- 3) 3.6V
- 4) 1.8V



Kirchhoff's laws and applications

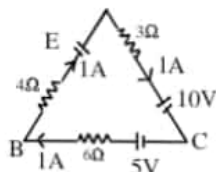
34. The value of current I, in the figure shown will be

- 1) 11A
- 2) 19A
- 3) 13A
- 4) 9A



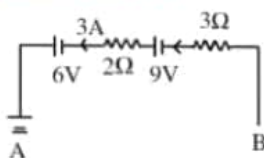
35. Find E for the Given Loop

- 1) -24v
- 2) 18v
- 3) 4v
- 4) 0v



36. The potential difference between points A and B is

- 1) 10v
- 2) 12v
- 3) 14v
- 4) 0v

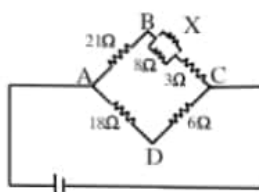


37. Four Resistances P, Q, R, X formed a wheat stone bridge. The bridge is balanced when $R=100\Omega$. If P and Q are inter changed the bridge balance for 121Ω . The value of x is

- 1) 100Ω
- 2) 200Ω
- 3) 300Ω
- 4) 110Ω

38. In the circuit shown in the figure, the value of Resistance X, when potential difference between the points B and D is zero will be

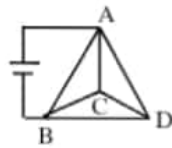
- 1) 9Ω
- 2) 8Ω
- 3) 6Ω
- 4) 4Ω



39. A 6Ω resistance is connected in the left gap of a meter bridge. In the second gap 3Ω and 6Ω are joined in parallel. The balance point of the bridge is at ____

- 1) 75cm
- 2) 60cm
- 3) 30cm
- 4) 25cm

40. When an unknown resistance and a resistance of 4Ω are used in the left and right gaps of a meter bridge, the balance point is 50cm. the shift in the balance point if a 4Ω resistance is now connected parallel to the resistor in right gap ____
- 1) $\frac{100}{3}$ cm 2) $\frac{50}{3}$ cm 3) $\frac{300}{3}$ cm 4) $\frac{400}{5}$ cm
41. When a conducting wire is connected in the left gap and known resistance in the right gap, the balance length is 75cm. If the wire is cut into 3 equal parts and one part is connected in the left gap, the balance length
- 1) shifts left by 25cms 2) shifts right by 25cms
3) shifts left by 50cms 4) shifts right by 50cms
42. A tetrahedron is consisting of 6 identical wires as shown in figure. Each wire is having a resistance of 2Ω . When an ideal cell of emf 2V is connected across AB, as shown then the current through CD is



- 1) 1 A 2) $\frac{6}{19}$ A 3) 4A 4) Zero

Potentiometer, Ammeter and voltmeter

43. The emf of a battery A is balanced by a length of 80cm on a potentiometer wire. The emf of a standard cell 1v is balanced by 50cm. The emf of A is
- 1) 2v 2) 1.4v 3) 1.5v 4) 1.6v
44. The balancing lengths of potentiometer wire are 800cm and 600cm when two cells of emf's E_1 and E_2 are connected in the secondary circuit first in series and then terminals of one cell is reversed; $\frac{E_1}{E_2}$ is equal to ____
- 1) 1/11 2) 14/11 3) 7/1 4) 4/3
45. When 6 identical cells of no internal resistance are connected in series in the secondary circuit of a potentiometer, the balancing length is 'l', balancing length becomes l/3 when some cells are connected wrongly, the number of cells connected wrongly are
- 1) 1 2) 3 3) 2 4) 4
46. In a potentiometer experiment, the balancing length with a cell is 560 cm. When an external resistance of 10 ohms is connected in parallel to the cell the balancing length changes by 60 cm. The internal resistance of the cell in ohms is :
- 1) 3.6 2) 2.4 3) 1.2 4) 0.6
47. The resistivity of a potentiometer wire is, if the area of cross section of the wire is 4cm^2 . The current flowing in the circuit is 1A, the potential gradient is 7.5 v/m ____
- 1) $3 \times 10^{-3} \Omega\text{-m}$ 2) $2 \times 10^{-6} \Omega\text{-m}$
3) $4 \times 10^{-2} \Omega\text{-m}$ 4) $5 \times 10^{-4} \Omega\text{-m}$

48. The length of potentiometer wire is 200 cm and the emf of standard cell in primary circuit is E volts. It is employed to a battery of emf 0.4 v the balance point is obtained at $l = 40\text{cm}$ from positive end, the E of the battery is (cell in primary is ideal and series resistance is zero)
- 1) 4v 2) 2v 3) 5v 4) 7v
49. The emf of a cell is 2V and its internal resistance is 2 ohm. A resistance of 8 ohm is joined to battery in parallel. This is connected in secondary circuit of potentiometer. If 1V standard cell balances for 100cm of potentiometer wire, the balance point of above cell is
- 1) 120cm 2) 240cm 3) 160cm 4) 116cm
50. A galvanometer has coil of resistance 50Ω and shows full deflection at $100\mu\text{A}$. The resistance to be added for the galvanometer to work as an ammeter of range 10mA is nearly
- 1) 0.5Ω in series 2) 0.5Ω in parallel
3) 5.0Ω in series 4) 5.0Ω in parallel
51. A galvanometer of resistance 1000Ω gives full-scale deflection for a current of 1mA. To measure a P.D of 10V, the resistance to be connected with the galvanometer is
- 1) $9\text{k}\Omega$ in series 2) $9\text{k}\Omega$ in parallel
3) $10\text{k}\Omega$ in series 4) $1\text{k}\Omega$ in parallel.
52. A galvanometer of 25 ohm resistance can read a maximum current of 6mA. It can be used as a voltmeter to measure maximum potential difference of 6V by connecting a resistance to galvanometer. Identify the correct choice from the following
- 1) 1025Ω in series 2) 1025Ω in parallel
3) 975Ω in series 4) 975Ω in parallel
53. A galvanometer has a current range of 15mA and voltage range of 750mv. To convert this galvanometer into an ammeter of range 25A, the shunt resistance required is nearly
- 1) 0.2Ω 2) 0.02Ω 3) 0.03Ω 4) 0.5Ω
54. In a galvanometer, a current of $1\mu\text{A}$ produces a deflection of 20 divisions. It has a resistance of 10Ω . If the galvanometer has 50 divisions on its scale and a shunt of 2.5Ω is connected across the galvanometer, the maximum current that the Galvanometer can measure now is
- 1) $12.5\mu\text{A}$ 2) 12.5 mA 3) $12.5 \times 10^{-7}\text{A}$ 4) $12.5 \times 10^{-3}\text{mA}$
55. Figures of merit of two galvanometers of resistance 100Ω and 50Ω , are 10^{-8}A/div and $2 \times 10^{-5}\text{A/div}$ respectively. In which case the voltage sensitivity is more ?
- 1) more in case I 2) more in case II
3) same in both cases 4) can't say
56. When 0.005A current flows through a moving coil galvanometer, it gives fullscale deflection. It is converted into a voltmeter to read 5 Volt, using an external resistance of 975Ω . The resistance of galvanometer in ohms is
- 1) 5 2) 10 3) 15 4) 25

57. A galvanometer of resistance $40\ \Omega$ and current passing through it is $100\ \mu\text{A}$ per division. The full scale has 50 divisions. If it is converted into an ammeter of range 2A by using a shunt, then the resistance of ammeter is
- 1) $\frac{40}{399}\ \Omega$ 2) $\frac{4}{399}\ \Omega$ 3) $0.01\ \Omega$ 4) $0.4\ \Omega$
58. An electrical meter of internal resistance $20\ \Omega$ gives a full scale deflection when one milliampere current flows through it. The maximum current, that can be measured by using three resistors of resistance $12\ \Omega$ each, in milliamperes is:
- 1) 10 2) 8 3) 6 4) 4
59. A student finds the balancing length to be 'l' with a cell of constant emf in the secondary circuit. Another student connects the same cell in the secondary circuit of potentiometer of double the length but with a cell of half the emf in the primary circuit. The balancing length will be [cell in primary is ideal and no series resistance is present in primary circuit]
- 1) 4l 2) l/4 3) 2l 4) l/2
60. A battery of negligible internal resistance is connected with 10m long wire. A standard cell gets balanced on 6m length of this wire. On increasing the length of potentiometer wire by 2m, the null point with the same standard cell in the secondary will be (no series resistance in primary)
- 1) increased by 2m 2) decreased by 2m
3) increased by 1.2m 4) decreased by 1.2m
61. In an experiment for calibration of voltmeter, a standard cell of emf 1.5V is balanced at 300cm length of potentiometer wire. The P.D. across a resistance in the circuit is balanced at 1.25m. If a voltmeter is connected across the same resistance, it reads 0.65V. The error in the volt meter is
- 1) 0.05V 2) 0.025V 3) 0.5V 4) 0.25V
62. A potentiometer wire of length 10m and resistance 30 ohm is connected in series with a battery of emf 2.5V, internal resistance 5 ohm and external resistance R. If the fall of potential along the potentiometer wire is 50mV/m, the value of R in ohms is
- 1) 115 2) 80 3) 50 4) 100

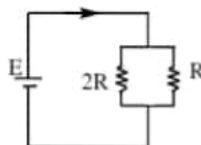
Electric power - Joules law

63. 60W-220V and 100W-220V bulbs are connected in series to 220V mains. The power consumption is more in the bulb of :
- 1) 60W 2) 100W
3) same for both the bulbs 4) none
64. 60 W, 100 W and 1000 W bulbs are connected in parallel to the mains. The bulb that will be brightest is (all of them have same specified voltage)
- 1) 60W 2) 100W 3) 1000W 4) all are equally bright
65. A bulb of 100W - 250 V is connected to 250 V mains. If another similar bulb is connected to it in series, the total power consumption is :
- 1) 100 W 2) 200 W 3) 50 W 4) 400 W

66. The power of a heating coil is P . It is cut into two equal parts. The power of one of them across same mains is :
 1) P 2) $2P$ 3) $4P$ 4) $P/2$
67. In a house there are four bulbs each of $50W$ and 5 fans each of $60W$. If they are used at the rate of 6 hours a day, the electrical energy consumed in a month of 30 days is
 1) 90000 units 2) 90 units 3) 0.9 units 4) 900 units
68. An electric kettle has two coils. When one coil is switched on it takes 15 minutes and the other takes 30 minutes to boil certain mass of water. The ratio of times taken by them, when connected in series and in parallel to boil the same mass of water is :
 1) $1 : 3$ 2) $5 : 2$ 3) $7 : 3$ 4) $9 : 2$
69. Two electric bulbs whose resistances are in the ratio of $1 : 2$ are connected in parallel to a constant voltage source. The powers dissipated in them have the ratio
 1) $1 : 2$ 2) $1 : 1$ 3) $2 : 1$ 4) $1 : 4$
70. A $10V$ storage battery of negligible internal resistance is connected across a 50Ω resistor. How much heat energy is produced in the resistor in 1 hour
 1) 7200J 2) 6200J 3) 5200J 4) 4200J

71. What is the ratio of heat generated in R and $2R$

- 1) $2 : 1$
 2) $1 : 2$
 3) $4 : 1$
 4) $1 : 4$

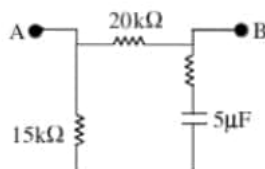


72. A $5^\circ C$ rise in the temperature is observed in a conductor by passing some current. When the current is doubled, then rise in temperature will be equal to
 1) $5^\circ C$ 2) $10^\circ C$ 3) $20^\circ C$ 4) $40^\circ C$

R.C. Circuits

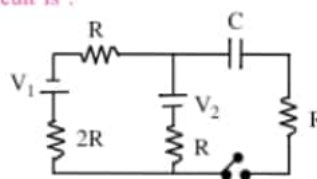
73. The value of the resistance as measured across terminals A and B in fig would be at steady state

- 1) $5k\Omega$
 2) $10k\Omega$
 3) $15k\Omega$
 4) $20k\Omega$



74. In the transient circuit shown, the time constant of the circuit is :

- 1) $\frac{5}{3}RC$ 2) $\frac{5}{2}RC$
 3) $\frac{7}{4}RC$ 4) $\frac{7}{3}RC$



Numerical Value type Questions

75. In a hydrogen atom, an electron is revolving with an angular frequency 6.28 rad/s around the nucleus. Then the equivalent electric current is 10^{-19} A
76. A copper wire of diameter 1 mm carries a current of 1.1 A . The drift speed of electrons is (given, density of $\text{Cu} = 9 \text{ g/cm}^3$, atomic weight of $\text{Cu} = 63 \text{ g}$ and one electron is contributed by each Cu atom) (in mm/s)
77. The temperature coefficient of resistivity of material is $0.0004 / \text{K}$. When the temperature of the material is increased by 50°C , its resistivity increases by $2 \times 10^{-8} \text{ ohm-m}$. The initial resistivity of the material in ohm-m is $\times 10^{-8}$
78. The emf of a Daniel cell is 1.08 V . When the terminals of the cells are connected to a resistance of 3Ω , the potential difference across the terminals is found to be 0.6 V . Then the internal resistance of the cell is Ω .
79. A 6 V cell with 0.5Ω internal resistance and 10 V cell with 1Ω internal resistance and a 12Ω external resistance are connected in parallel. The current in amp through the 10 V cell is
80. In a meter bridge experiment Two unknown resistances X and Y are connected to left and right gaps of a meter bridge and the balancing point is obtained at 20 cm from left ($X < Y$) The new position of the null point from left if one decides balance a resistance of $4X$ against Y . (in cm)
81. How many time constants will elapse before the current in a charging RC circuit drops to half of its initial value? Answer the same question for a discharging RC circuit?
82. How many time constants will elapse before the energy stored in the capacitor reaches half of its equilibrium value in a charging RC circuit?
83. How many time constants will elapse before the power delivered by the battery drops to half of its maximum value in an RC circuit?

KEY SHEET

EXERCISE-I

1) 1	2) 2	3) 2	4) 3	5) 4	6) 3	7) 3	8) 3	9) 2	10) 1
11) 4	12) 2	13) 4	14) 3	15) 1	16) 3	17) 1	18) 4	19) 3	20) 2
21) 1	22) 2	23) 3	24) 3	25) 1	26) 3	27) 1	28) 3	29) 1	30) 4
31) 2	32) 3	33) 4	34) 1	35) 2	36) 3	37) 1	38) 1	39) 2	40) 2
41) 2	42) 2	43) 3	44) 4	45) 1	46) 3	47) 4	48) 4	49) 3	50) 3
51) 4	52) 2	53) 3	54) 2	55) 2	56) 1	57) 2	58) 4	59) 4	60) 4
61) 3	62) 2	63) 1	64) 2	65) 1	66) 4	67) 4	68) 3	69) 3	70) 1
71) 0.16	72) 46.67	73) 15	74) 20	75) 1.95	76) 4.8	77) 79.995	78) 192	79) 20	
80) 0.36									

EXERCISE-II

- | | | | | | | | | | |
|----------------|----------|----------|-------|---------|---------|---------|---------|----------|--------|
| 1) 1 | 2) 4 | 3) 1 | 4) 2 | 5) 2 | 6) 3 | 7) 3 | 8) 1 | 9) 4 | 10) 1 |
| 11) 4 | 12) 2 | 13) 3 | 14) 3 | 15) 3 | 16) 1 | 17) 2 | 18) 4 | 19) 3 | 20) 3 |
| 21) 2 | 22) 2 | 23) 3 | 24) 3 | 25) 2 | 26) 3 | 27) 4 | 28) 4 | 29) 2 | 30) 2 |
| 31) 2 | 32) 3 | 33) 4 | 34) 1 | 35) 2 | 36) 4 | 37) 4 | 38) 2 | 39) 1 | 40) 2 |
| 41) 1 | 42) 4 | 43) 4 | 44) 3 | 45) 3 | 46) 3 | 47) 1 | 48) 2 | 49) 3 | 50) 2 |
| 51) 1 | 52) 3 | 53) 2 | 54) 1 | 55) 1 | 56) 4 | 57) 1 | 58) 3 | 59) 1 | 60) 3 |
| 61) 2 | 62) 1 | 63) 1 | 64) 3 | 65) 3 | 66) 2 | 67) 2 | 68) 4 | 69) 3 | 70) 1 |
| 71) 1 | 72) 3 | 73) 4 | 74) 3 | 75) 1.6 | 76) 0.1 | 77) 100 | 78) 2.4 | 79) 2.87 | 80) 50 |
| 81) 0.69, 0.69 | 82) 1.23 | 83) 0.69 | | | | | | | |